Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

1. What is the basic concept of similarity in data mining? 1 / 1 point

- It measures the complexity of a data mining algorithm.
- It measures how much two data objects are alike.
- It measures the speed of data processing.
- It measures the size of a dataset.

Nice work

Correct! Similarity measures how much two data objects are alike, which is fundamental in data mining.

2. What is the relationship between similarity and distance in data mining? 1 / 1 point

- Similarity measures likeness, while distance measures dissimilarity.
- Similarity and distance are unrelated concepts.
- Both measure how complex a dataset is.
- Both are used to measure the speed of data processing.

Nice work

Correct! Similarity measures how alike two objects are, while distance measures how dissimilar they are.

3. What does the intersection of two itemsets represent? 1 / 1 point

- The difference between the two itemsets. Please revisit the Similarity of Itemsets lesson materials before attempting this quiz again.
- The largest common subset of the two itemsets.
- The total number of items in both itemsets.
- The smallest common superset of the two itemsets.

Nice work

Correct! The intersection is the largest common subset of the two itemsets.

4. Why might longer itemsets appear more similar than they actually are? 1 / 1 point

- Because they contain more unique items.
- Because they are more expensive.
- Because they are more complex to analyze.
- Because they are more likely to share items with other itemsets.

Nice work

Correct! Longer itemsets are more likely to share items with others, which can falsely indicate higher similarity.

5. What is the union of two itemsets? 1 / 1 point

- The difference between the two itemsets.
- The largest common subset of the two itemsets.
- The total number of unique items in both itemsets.
- The smallest common superset of the two itemsets.

Nice work

Correct! The union is the smallest common superset of the two itemsets.

6. What is the Jaccard similarity index used for? 1 / 1 point

- To determine the probability of an event occurring.
- To find the average of a set of numbers.
- To calculate the distance between two points in a plane.
- To measure the similarity between two item sets.

Nice work

Correct! The Jaccard similarity index measures the similarity between two item sets by comparing their intersection and union.

7. How is the Jaccard similarity between two item sets calculated? 1 / 1 point

- By subtracting the number of items in the union from the number of items in the intersection.
- By adding the number of items in the intersection and the union.
- By multiplying the number of items in the intersection by the number of items in the union.
- By dividing the number of items in the intersection by the number of items in the union.

Nice work

Correct! The Jaccard similarity is calculated by dividing the size of the intersection by the size of the union of the two sets.

8. What does a Jaccard similarity index of 1 indicate? 1 / 1 point

- The two item sets have a large union.
- The two item sets are identical.
- The two item sets have a small intersection.
- The two item sets have no items in common.

Nice work

Correct! A Jaccard similarity index of 1 indicates that the two item sets are identical.

9. What is the Jaccard distance? 1 / 1 point

- The product of the Jaccard similarity and the union size.
- The sum of the Jaccard similarity and the intersection size.
- 1 minus the Jaccard similarity.
- The difference between the union size and the intersection size.

Nice work

Correct! The Jaccard distance is calculated as 1 minus the Jaccard similarity.

10. Which property does the Jaccard similarity index have? 1 / 1 point

- Asymmetric
- Transitive
- Symmetric
- Reflexive

Nice work

Correct! The Jaccard similarity index is symmetric, meaning the similarity from set A to B is the same as from B to A.

11. How can items be transformed to measure similarity using Jaccard index? 1 / 1 point

- By grouping items into categories.
- By representing each item as a set of transactions in which it appears.
- By listing each item in alphabetical order.
- By converting each item into a numerical value.

Nice work

Correct! Items can be transformed into sets of transactions they appear in, allowing similarity measurement using the Jaccard index.

12. What is an example of applying the Jaccard similarity in text mining? 1 / 1 point

- Spelling correction
- Grammar checking
- Calculating word frequency
- Plagiarism detection

Nice work

Correct! Jaccard similarity can be used in text mining for plagiarism detection by comparing document similarities.

13. What does a Jaccard similarity index of 0 indicate? 1 / 1 point

- The two item sets are identical.
- The two item sets have a large intersection.
- The two item sets have a small union.
- The two item sets share no items.

Nice work

Correct! A Jaccard similarity index of 0 indicates that the two item sets have no items in common.

14. What is the Jaccard similarity between two identical item sets? 1 / 1 point

- 2
- 0
- 1
- 0.5

Nice work

Correct! The Jaccard similarity between two identical item sets is 1, as their intersection and union are the same.

15. What is the potential use of Jaccard similarity in market segmentation? 1 / 1 point

- To calculate sales tax.
- To determine product prices.
- To manage inventory levels.
- To group customers with similar shopping baskets.

Nice work

Correct! Jaccard similarity can be used to group customers with similar shopping baskets for market segmentation.